CSC 710 — Software Engineering

Project 1: Battleship Online Multiplayer

Final Project Report

Course: CSC 710 — Software Engineering **Instructor:** Dr. Tianxiao Zhang **Date:** March 4, 2026 **Team Members:**

- Merve Gazi (Product Owner)
- Umut Celik (Developer)
- Justin Huang (Developer)

Repository: GitHub — CSC-710-Battleship-Online-Multiplayer **Live Demo:** GitHub Pages

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1. Project Overview

1.1 Problem Statement

Traditional board games like Battleship are limited to in-person, two-player sessions. This project aims to bring the classic naval combat experience online, allowing players to compete in real-time from any device with a web browser.

1.2 Project Description

A real-time, browser-based multiplayer Battleship game where players can join a lobby, find opponents through FIFO matchmaking or custom tables, place ships on a 10×10 grid, and battle in turn-based combat. The application features user authentication, live lobby status with presence tracking, in-lobby text chat, player profiles with win/loss statistics, and a responsive UI that works across desktop and mobile devices.

1.3 Development Approach

This is a **Greenfield Development** project — built entirely from scratch with no legacy code or existing infrastructure. The team followed **Agile Scrum** methodology with three one-week sprints spanning February 11 to March 4, 2026.

1.4 Classic Battleship Rules

Ship	Size (cells)
Carrier	5
Battleship	4
Cruiser	3
Submarine	3
Destroyer	2

- Grid: 10×10 (columns A–J, rows 1–10)
- Ships placed horizontally or vertically, no overlap, within bounds
- Players alternate turns, selecting one cell on the opponent's grid
- Result: **Hit** (ship segment) or **Miss** (empty water)
- Ship is **Sunk** when all its cells are hit
- Game ends when all five opponent ships are sunk

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2. Team Organization & Scrum Roles

2.1 Team Structure

The team adopted a flat Scrum structure. Merve Gazi served as **Product Owner**, responsible for backlog prioritization and task breakdown. All three members functioned as **Developers**, each independently picking tasks from the backlog, implementing them, and committing their work.

Member	Scrum Role	Primary Responsibilities
Merve Gazi	Product Owner / Developer	Backlog management, matchmaking logic, board UI, presence system, responsive design, ship visuals
Umut Celik	Developer	Auth system, lobby UI, routing, custom tables, attack mechanism, realtime sync, game-end stats, PR reviews
Justin Huang	Developer	Project setup (React/Vite/Tailwind), Supabase configuration, ship placement validation, game mechanics enforcement

2.2 Communication

- **Week 1:** In-class face-to-face meeting (~30-40 minutes) during lecture time
- **Ongoing:** Natural discussion for coordination; Justin had domain expertise in Battleship game mechanics, Merve was familiar with the game rules, Umut contributed on technology selection and architecture decisions
- **Code Reviews:** Pull requests on GitHub with review before merging to main

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3. Requirements Engineering

3.1 Requirements Gathering

Requirements were gathered during the first week through an in-class face-to-face meeting. The team used **natural discussion** as the primary brainstorming technique. Justin's existing familiarity with Battleship game mechanics guided the game rules definition, while Umut contributed technology stack decisions (React + Supabase). The team collectively refined requirements into a comprehensive Technical Design Document (`CSC710_Battleship_Technical_Document.md`).

Stakeholders:

- Dr. Tianxiao Zhang (Instructor / Primary Stakeholder)
- Team members (Developers and Users)
- Classmates (Potential end users for demo)

3.2 User Stories

ID	User Story	Priority	Sprint
US-001	As a player, I want to register with my email so that I can have a persistent account	P0	1
US-002	As a player, I want to log in so that I can access the lobby	P0	1
US-003	As a player, I want to see who is online in the lobby so that I can find opponents	P0	1
US-004	As a player, I want to send chat messages in the lobby so that I can communicate with others	P1	1
US-005	As a player, I want to click “Quick Match” so that I can be automatically paired with an opponent	P0	2
US-006	As a player, I want to create a custom table so that I can invite a specific opponent	P0	2
US-007	As a player, I want to request to join another player’s table so that I can challenge them	P0	2
US-008	As a player, I want to place my 5 ships on a 10×10 grid using drag-and-drop so that I can prepare for battle	P0	2
US-009	As a player, I want to rotate ships before placing them so that I can choose horizontal or vertical orientation	P0	2
US-010	As a player, I want to attack a cell on my opponent’s grid so that I can try to hit their ships	P0	2
US-011	As a player, I want to see hit/miss/sunk feedback immediately so that I know my attack result	P0	2
US-012	As a player, I want to see a game-over screen with stats when all ships are sunk so that I know who won	P0	3
US-013	As a player, I want my win/loss record updated after each game so that I can track my progress	P1	3
US-014	As a player, I want the game to detect if my opponent disconnects so that I’m not stuck waiting	P1	3
US-015	As a player, I want the game to work on my phone so that I can play on mobile	P1	3

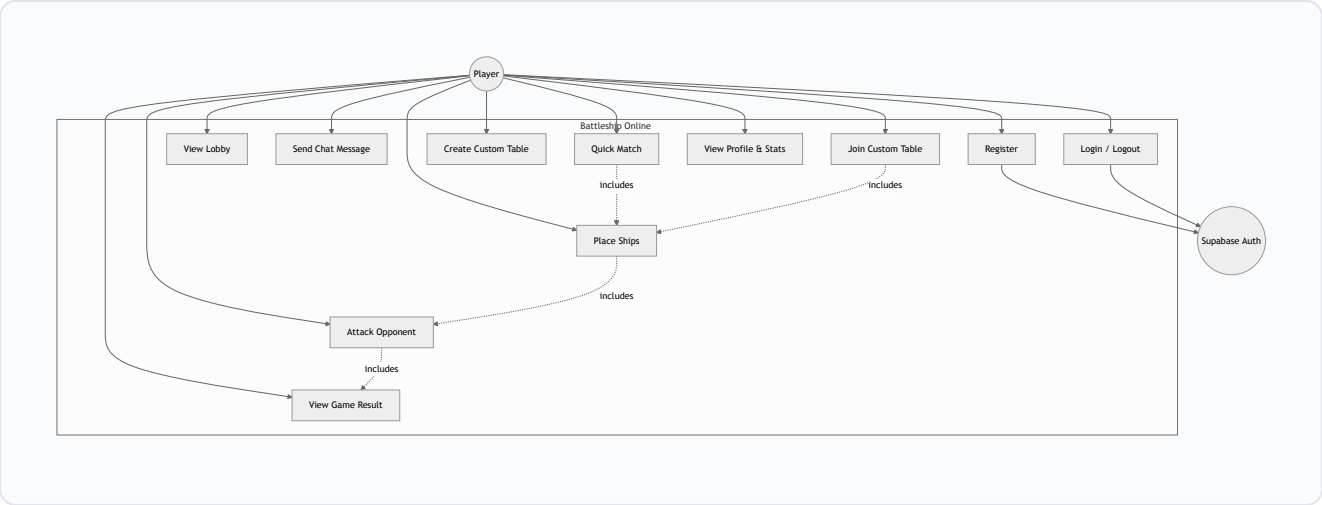
3.3 Requirements Stack

REQ ID	Description	Story Points	Priority	Sprint
REQ-001	User registration with email/password via Supabase Auth	2	P0	1
REQ-002	User login with session management and JWT tokens	2	P0	1
REQ-003	Protected routes — redirect unauthenticated users to login	1	P0	1
REQ-004	Lobby page with real-time online user list (Supabase Presence)	3	P0	1
REQ-005	Lobby chat with real-time message sync	3	P1	1
REQ-006	Landing page with “Play Now” call-to-action	1	P0	1
REQ-007	FIFO matchmaking queue with mutual-accept handshake	5	P0	2
REQ-008	Custom table creation with join-request approval flow	5	P0	2
REQ-009	10×10 game board grid component (responsive)	3	P0	2
REQ-010	Ship placement with drag-and-drop, rotation, and validation	8	P0	2
REQ-011	Attack mechanism with hit/miss/sunk resolution	5	P0	2
REQ-012	Turn-based gameplay with real-time move synchronization	5	P0	2
REQ-013	Game-end detection (all ships sunk) and result screen	3	P0	3
REQ-014	Player profile page with win/loss/total games stats	2	P1	3

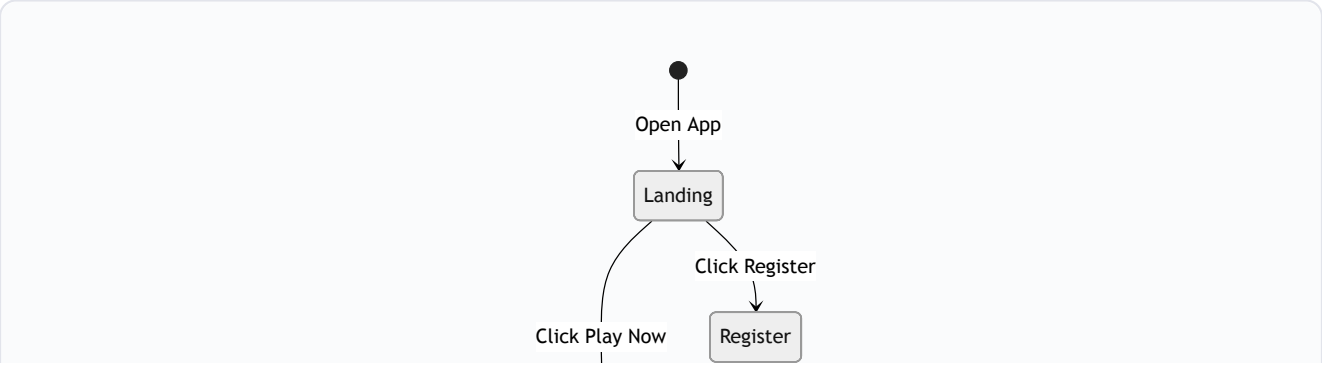
REQ-015	Heartbeat-based disconnect detection (10s interval, 30s threshold)	3	P1	3
REQ-016	Mobile responsive layout with board tab switcher	3	P1	3
REQ-017	Ship SVG visuals for realistic appearance	2	P2	3
REQ-018	CI/CD pipeline with GitHub Actions → GitHub Pages	2	P0	1
	Total	**57**		

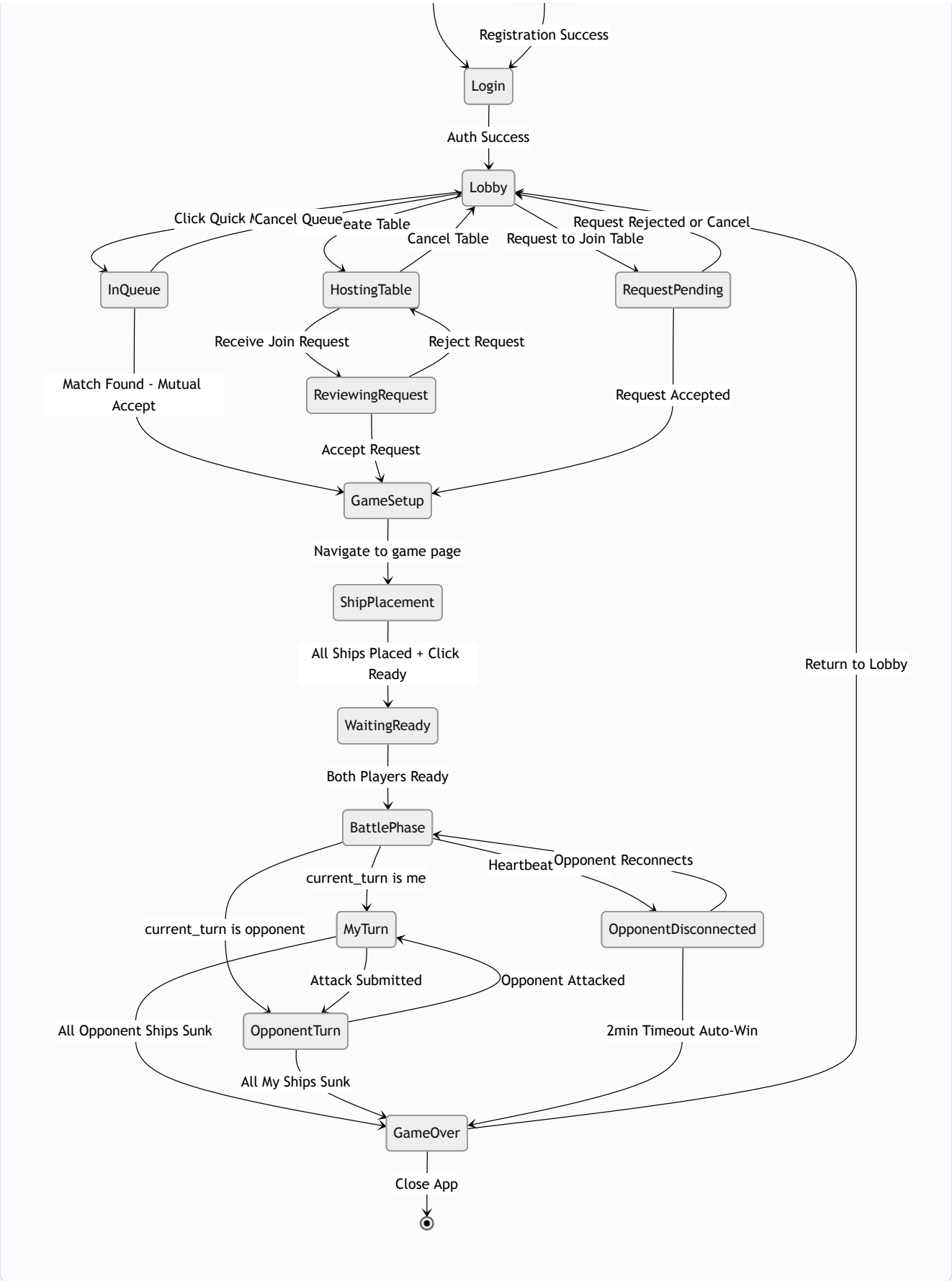
3.4 UML Diagrams

3.4.1 Use Case Diagram

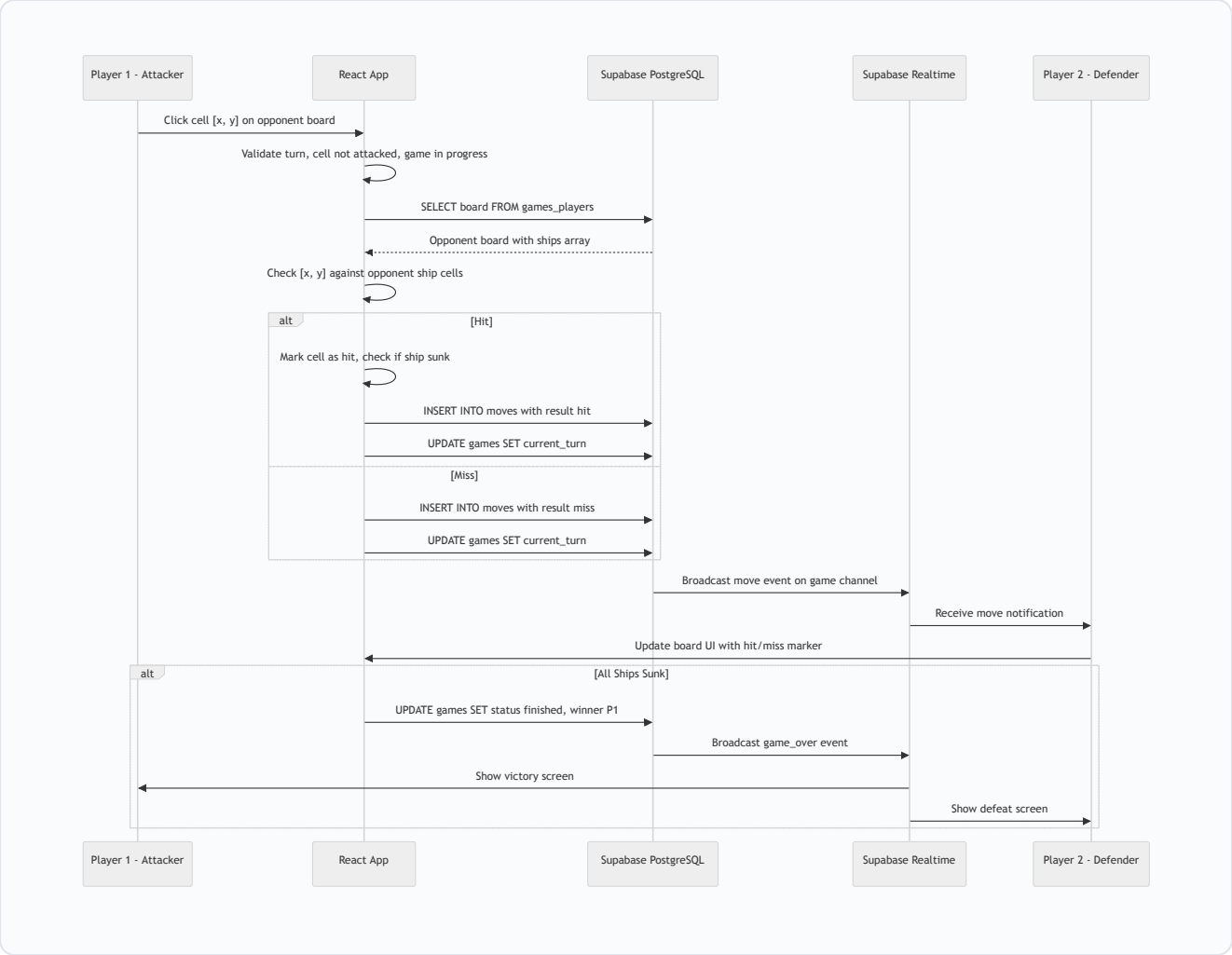


3.4.2 Game State Machine Diagram




Registration Success
Login
Auth Success
Lobby
Click Quick M
Cancel Queue
eate Table
Cancel Table
Request to Join Table
Request Rejected or Cancel
InQueue
HostingTable
RequestPending
Receive Join Request
Reject Request
Match Found - Mutual Accept
ReviewingRequest
Request Accepted
Accept Request
GameSetup
Navigate to game page
ShipPlacement
All Ships Placed + Click Ready
Return to Lobby
WaitingReady
Both Players Ready
BattlePhase
Heartbeat
Opponent Reconnects
current_turn is me
current_turn is opponent
MyTurn
OpponentDisconnected
Attack Submitted
Opponent Attacked
All Opponent Ships Sunk
OpponentTurn
2min Timeout Auto-Win
All My Ships Sunk
GameOver
Close App

3.4.3 Sequence Diagram — Attack Flow



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4. System Architecture & Design

4.1 Technology Stack

```
Frontend:    React 18 + TypeScript + Vite 6 + Tailwind CSS v4
Backend:     Supabase (PostgreSQL + Auth + Realtime WebSocket)
Hosting:     GitHub Pages (static SPA)
CI/CD:       GitHub Actions (auto-deploy on push to main)
```

4.2 Design Rationale

Decision	Choice	Rationale
Frontend Framework	React 18	Industry standard, large ecosystem, team familiarity
Language	TypeScript	Type safety for complex game state, better DX with IDE support
Build Tool	Vite 6	Fast HMR, native ESM, simpler config than Webpack
CSS Framework	Tailwind CSS v4	Utility-first, rapid UI development, responsive design built-in
Backend	Supabase	PostgreSQL + Auth + Realtime in one platform, generous free tier, no custom server needed
Hosting	GitHub Pages	Free, integrates with GitHub Actions, sufficient for static SPA
State Management	React Context + useReducer	Lightweight, no external dependency, sufficient for this app's complexity
Routing	React Router v7	De facto standard for React SPAs

4.3 Architecture Diagram

```
┌──────────────────────────────────────────────────────────────┐
│                      Client (Browser)                        │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐  ┌──────────┐     │
│  │ Landing  │  │ Auth     │  │ Lobby    │  │ Game     │     │
│  │ Page     │  │ Pages    │  │ Page     │  │ Page     │     │
│  └──────────┘  └──────────┘  └──────────┘  └──────────┘     │
│      │             │            │             │             │
│  ┌──────────────────────────────────────────────────────┐   │
│  │          React Context (Auth + Game State)           │   │
│  └──────────────────────────────────────────────────────┘   │
│      │             │            │             │             │
│  ┌──────────────────────────────────────────────────────┐   │
│  │             Supabase Client SDK                      │   │
│  └──────────────────────────────────────────────────────┘   │
└──────────────────────────┬──────────────────────────────────┘
                           │ HTTPS + WebSocket
┌──────────────────────────┴──────────────────────────────────┐
│                      Supabase Cloud                          │
│  ┌──────────┐  ┌──────────┐  ┌──────────────────────────┐   │
│  │   Auth   │  │ Realtime │  │     PostgreSQL           │   │
│  │  (JWT)   │  │ (WS)     │  │  profiles, games,        │   │
│  │          │  │          │  │  games_players, moves,   │   │
│  │          │  │          │  │  tables, chat_messages,  │   │
│  │          │  │          │  │  matchmaking_queue       │   │
│  └──────────┘  └──────────┘  └──────────────────────────┘   │
└──────────────────────────────────────────────────────────────┘
```

4.4 Realtime Channel Architecture

Channel	Purpose
`lobby` (Presence)	Track online users and their status
`lobby-chat`	Chat message inserts
`lobby-tables`	Table create/update/delete
`table:{tableId}`	Join requests for specific table
`matchmaking`	Queue changes, match notifications
`game:{gameId}`	Moves, ready status, game state changes

4.5 Component Tree

```
App
├── LandingPage
├── LoginPage
├── RegisterPage
├── ProtectedRoute
│   ├── LobbyPage
│   │   ├── LobbyHeader
│   │   ├── LobbyStats
│   │   ├── OnlineUsersList
│   │   ├── QuickMatchButton
│   │   ├── CreateTableButton
│   │   ├── TableList
│   │   │   └── TableCard
│   │   ├── HostTableModal
│   │   │   └── RequestItem
│   │   └── LobbyChat
│   ├── GamePage
│   │   ├── TurnIndicator
│   │   ├── BoardGrid (x2: player + opponent)
│   │   │   └── BoardCell
│   │   └── GameEndModal
│   └── ProfilePage
├── AuthLayout
├── Modal
├── Button
├── Input
└── LoadingSpinner
```

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5. Database Design

5.1 Entity Relationship Diagram

```
┌──────────────┐        ┌──────────────┐        ┌──────────────┐
│   PROFILES   │        │    GAMES     │        │    MOVES     │
├──────────────┤        ├──────────────┤        ├──────────────┤
│ id (PK)      │──┐     │ id (PK)      │──┐     │ id (PK)      │
│ email (UK)   │  │     │ table_id(FK) │  │     │ game_id (FK) │
│ display_name │  │     │ status       │  ├────>│ player_id(FK)│
│ total_games  │  │     │ current_turn │  │     │ x, y         │
│ wins, losses │  │     │ winner_id    │  │     │ result       │
│ created_at   │  │     │ created_at   │  │     │ sunk_ship    │
└──────────────┘  │     └──────────────┘  │     │ move_number  │
                  │           │           │     └──────────────┘
                  │           │ 1:N       │
                  │     ┌──────────────┐  │
                  │     │GAMES_PLAYERS │  │
                  │     ├──────────────┤  │
                  ├────>│ game_id (FK) │<─┘
                  │     │ player_id(FK)│
                  │     │ board (JSONB)│
                  │     │ ready        │
                  │     │ player_number│
                  │     └──────────────┘
                  │
                  │     ┌──────────────┐        ┌────────────────┐
                  │     │   TABLES     │        │TABLE_REQUESTS  │
                  │     ├──────────────┤        ├────────────────┤
                  ├────>│ host_id (FK) │──1:N>│ table_id (FK)  │
                  │     │ status       │        │ requester_id   │
                  │     └──────────────┘        │ status         │
                  │                             └────────────────┘
                  │
                  │     ┌──────────────┐        ┌────────────────┐
                  ├────>│CHAT_MESSAGES │        │MATCHMAKING_QUE │
                  │     ├──────────────┤        ├────────────────┤
                  │     │ sender_id(FK)│        │ player_id (FK) │
                  │     │ message      │        │ joined_at      │
                  │     │ channel      │        └────────────────┘
                  └────>│ created_at   │
                        └──────────────┘
```

5.2 Key Design Decisions

- **Board state as JSONB:** Ships array stored in `games_players.board` allows flexible board representation without a separate ships table
- **Move history table:** Every attack is recorded for potential replay functionality
- **Matchmaking queue:** Simple FIFO with `DELETE ... RETURNING` for atomic queue pop to prevent race conditions
- **Triggers:** Auto-create `profiles` row on `auth.users` insert via PostgreSQL trigger

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6. Implementation

6.1 Feature Implementation Status

Feature	Status	Sprint
Landing Page	Completed	1
User Registration (Supabase Auth)	Completed	1
User Login / Logout	Completed	1
Protected Routes	Completed	1
Lobby UI + Presence	Completed	1
Lobby Chat (Realtime)	Completed	1
CI/CD + GitHub Pages Deploy	Completed	1
Quick Match (FIFO Matchmaking)	Completed	2
Custom Tables + Join Requests	Completed	2
Ship Placement (Drag & Drop)	Completed	2
Attack Mechanism (Hit/Miss/Sunk)	Completed	2
Realtime Move Sync	Completed	2
Game End Detection + Stats	Completed	3
Profile Page (Win/Loss)	Completed	3
Heartbeat Disconnect Detection	Completed	3
Mobile Responsive + Tab Switcher	Completed	3
Ship SVG Visuals	Completed	3

All planned features were completed and deployed.

6.2 Technical Challenges & Solutions

Challenge	Solution
Matchmaking race condition — two players popping the same queue entry	Used `DELETE ... RETURNING` for atomic queue pop; added mutual-accept handshake
Realtime chat duplicates — messages appearing twice	Switched from polling to Supabase Realtime subscription; removed stale polling code
Move number race condition — concurrent moves causing constraint errors	Added `move_number` tracking with `upsert ignoreDuplicates`
Mobile board overflow — 10×10 grid exceeding viewport on small screens	Implemented tab switcher for "My Board" vs "Enemy Board" on mobile
Ghost matches — matchmaking detecting offline users as available	Added stale game filtering and offline player detection via Presence
Ship SVG displacement — SVG overlays pushing grid cells out of alignment	Rendered ships as single continuous SVG per ship with absolute positioning

6.3 Design by Contract — Key Functions

Following the Design by Contract principles from Lecture 5, critical game functions have clear preconditions and postconditions:

Ship Placement Validation (`gameLogic.ts`):

- *Precondition:* Exactly 5 ships with correct sizes, all cells within 0-9 bounds, no overlapping cells, cells are contiguous and match orientation
- *Postcondition:* Board state is valid and ready for battle; `games_players.ready` set to `true`
- *Invariant:* Total ship cells = 5 + 4 + 3 + 3 + 2 = 17

Attack Resolution (`useGame.ts`):

- *Precondition:* Game status is `in_progress`, `current_turn` matches attacker's ID, target cell not previously attacked
- *Postcondition:* Move recorded in `moves` table, `current_turn` switched to opponent, board UI updated
- *Invariant:* Move count strictly increases; each cell attacked at most once

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7. Open-Source Components

Component	Version	License	Purpose
React	18.x	MIT	UI framework
React DOM	18.x	MIT	DOM rendering
React Router	7.x	MIT	Client-side routing
TypeScript	5.x	Apache 2.0	Type-safe JavaScript
Vite	6.x	MIT	Build tool and dev server
Tailwind CSS	4.x	MIT	Utility-first CSS framework
@supabase/supabase-js	2.x	MIT	Supabase client SDK

All dependencies use permissive open-source licenses (MIT / Apache 2.0) with no licensing conflicts. No components were copy-pasted; all are installed via npm and imported through standard module resolution, following the **Adapter/Wrapper pattern** recommended in Lecture 6.

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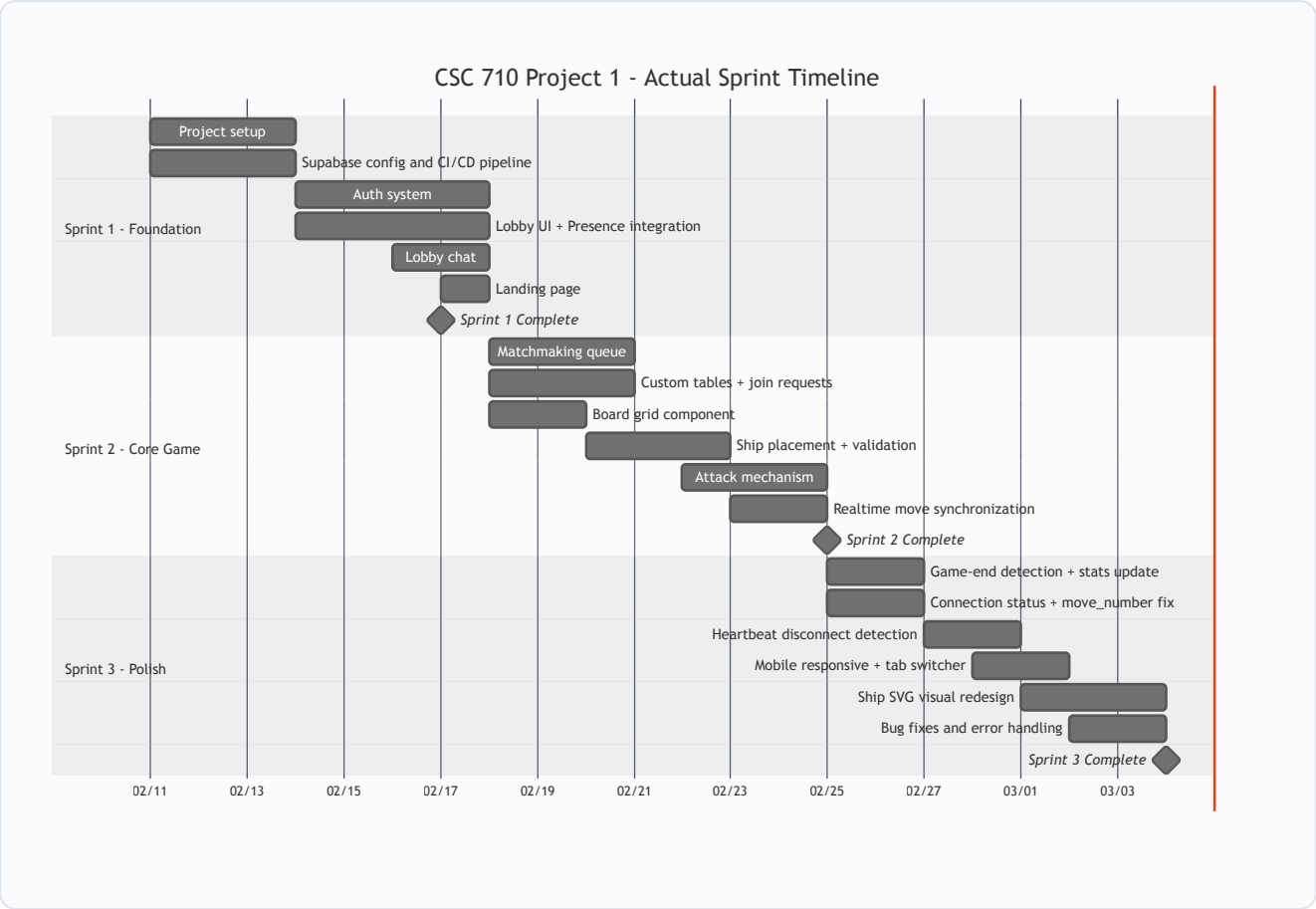
8. Sprint Management & Progress

8.1 Sprint Timeline



8.2 Sprint Backlog Summary

Sprint	Duration	Story Points	Key Deliverables
Sprint 1	Feb 11–17	14 SP	Auth, Lobby, Chat, CI/CD, Landing Page
Sprint 2	Feb 18–25	29 SP	Matchmaking, Custom Tables, Ship Placement, Attack, Realtime Sync
Sprint 3	Feb 25–Mar 4	14 SP	Game End, Profiles, Disconnect Detection, Mobile, Ship Visuals
Total	**3 weeks**	**57 SP**	**All features completed**

8.3 Velocity

Sprint	Planned SP	Completed SP	Velocity
Sprint 1	14	14	14 SP/week
Sprint 2	29	29	29 SP/week
Sprint 3	14	14	14 SP/week
Average			**19 SP/week**

Sprint 2 had the highest velocity as the core game mechanics required the most effort and all three team members were working in parallel on different subsystems.

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9. Team Contributions

9.1 Git Statistics

Team Member	Commits	Lines Added	Lines Deleted	Key Contributions
Merve Gazi	37	4,815	1,014	Matchmaking, board grid, presence, chat, responsive design, ship SVGs, heartbeat, mobile tab switcher
Umut Celik	26	3,736	474	Auth system, routing, lobby UI, custom tables, attack mechanism, realtime sync, game-end stats, PR reviews
Justin Huang	10	4,621	615	Project setup (React/Vite/Tailwind/Supabase), ship placement validation, game mechanics enforcement, matchmaking fixes
Total	**73**	**13,172**	**2,103**	

9.2 Contribution Matrix by Feature

Feature	Merve	Umut	Justin
Project Setup & Config			X
Supabase Wiring & CI/CD			X
Auth (Login/Register)		X	
Routing & Protected Routes		X	
Lobby UI & Presence	X	X	
Lobby Chat	X	X	
Quick Match (Matchmaking)	X		X
Custom Tables		X	X
Board Grid Component	X		
Ship Placement & Validation			X
Attack Mechanism		X	
Realtime Move Sync		X	
Game End + Stats		X	
Heartbeat & Disconnect	X		
Mobile Responsive	X		
Ship SVG Visuals	X		
Landing Page		X	
Profile Page		X	

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10. Known Issues & Bug Fixes

10.1 Bugs Encountered & Resolved

Bug	Severity	Sprint	Resolution
Chrome autofilling display name as login username	Low	2	Added `autocomplete="off"` attribute
Chat messages appearing twice	Medium	2	Replaced polling with Realtime subscription
Matchmaking detecting offline ghost players	High	2	Added stale game filtering + Presence-based offline detection
Both players not seeing match notification	High	2	Added polling fallback for race condition + match info UI
Ship SVG overlays displacing grid cells	Medium	3	Used absolute positioning for SVG overlays
Board overflow on mobile screens	Medium	3	Implemented tab switcher for My Board / Enemy Board
`move_number` constraint error on duplicate moves	High	3	Used `upsert` with `ignoreDuplicates` option
Viewport scroll issues on game page	Medium	3	Restored full-size boards with internal scroll

10.2 Demo Day Issues

During the in-class demo, several error-handling edge cases surfaced under live conditions. These were identified and fixed in the final sprint, including lobby state cleanup on refresh and auto-dismissing stale notices.

10.3 Known Limitations

Limitation	Reason	Mitigation
No server-side move validation	Client-side only (academic project, no Supabase Edge Functions)	Client validates turn order, cell uniqueness, game status
No Row Level Security (RLS)	Development speed trade-off	Application-level access control; opponent board data is theoretically accessible via direct Supabase queries
No rate limiting on chat	No server-side enforcement	Client-side throttle (1 msg/sec), max 500 chars
Reconnect window is best-effort	Heartbeat-based detection, not guaranteed	10s heartbeat, 30s stale threshold, 2-min reconnect window

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11. Sprint Retrospective

Sprint 1 (Feb 11–17): Foundation

What Went Well	What Didn't Go Well
Tech stack decisions were made quickly and proved effective	Initial Supabase Realtime configuration had a learning curve
CI/CD pipeline was set up early, enabling fast iteration	

Sprint 2 (Feb 18–25): Core Game

What Went Well	What Didn't Go Well
Parallel development on different subsystems was efficient	Matchmaking race conditions required multiple iterations to fix
Pull request workflow caught issues before merging	Chat duplicate messages took debugging effort

Sprint 3 (Feb 25–Mar 4): Polish

What Went Well	What Didn't Go Well
Delivered a fully working, deployable product aligned with the technical design document	Some game mechanics were not defined in the original document and had to be updated during the final week
All planned features were completed	Demo day exposed error-handling edge cases that needed last-minute fixes

Overall Improvements for Project 2

- Define all game mechanics thoroughly in the design document before implementation begins
- Add automated tests early to catch regressions
- Set up more granular error handling from the start

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12. Future Work (Project 2)

Project 2 will extend this Battleship project as a **Brownfield Development** effort, adding the following features:

Feature	Description	Estimated SP
AI Opponent	Single-player mode against a computer opponent with varying difficulty levels	13
Custom Maps	Different grid sizes and map layouts for varied gameplay	8
Sound Effects	Audio cues for hit, miss, sunk, victory, and defeat	3
Enhanced Visuals	Explosion animations, water splash effects, improved ship graphics	5
Additional Game Mechanics	New ship types, power-ups, or special abilities	8

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13. Lessons Learned

1. **Supabase Realtime is powerful but has edge cases.** Race conditions in matchmaking and move synchronization required creative solutions like mutual-accept handshakes and upsert with `ignoreDuplicates`.

2. **Define game mechanics completely before coding.** Several mechanics (like one-ship-per-turn placement, mutual match acceptance) were not in the original design document and had to be added during implementation.

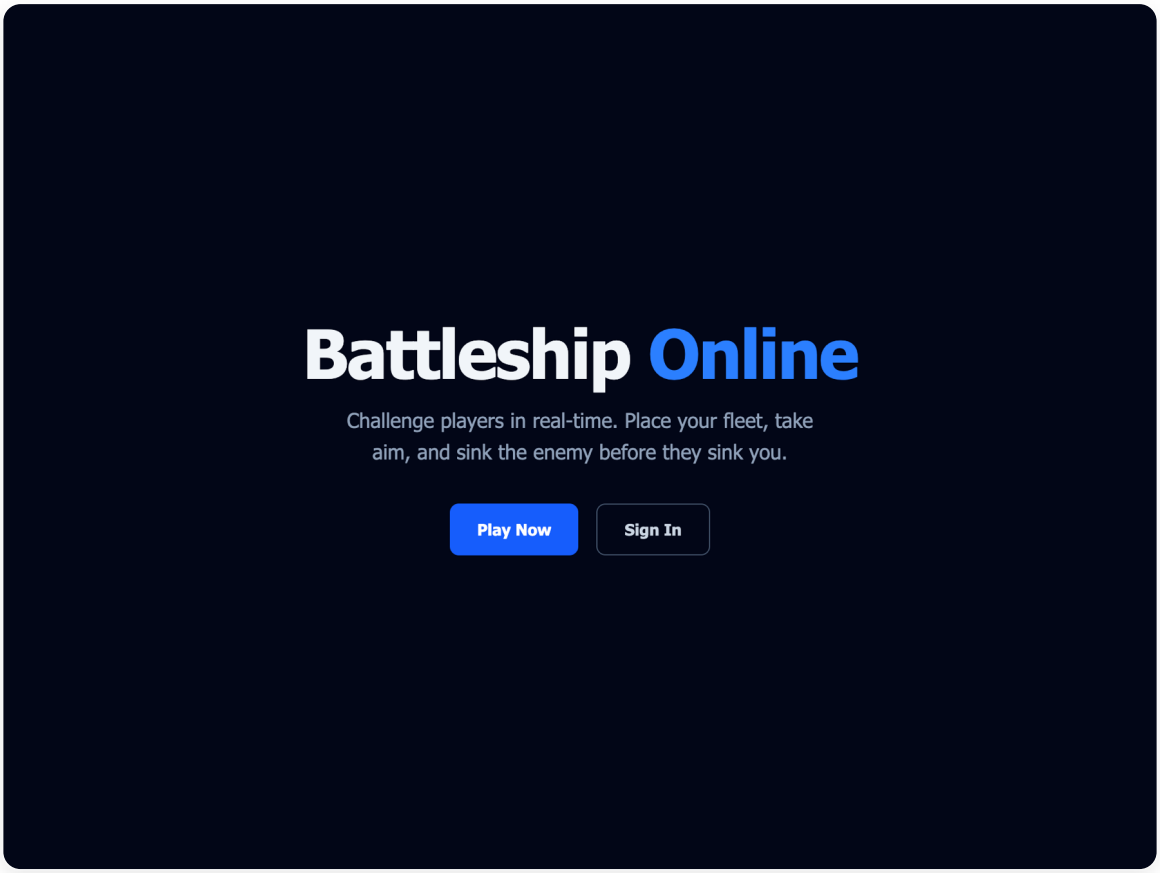
3. **Mobile-first design saves rework.** The 10×10 grid overflow issue on mobile required significant refactoring in Sprint 3. Starting with mobile constraints would have prevented this.

4. **CI/CD from day one pays off.** Having GitHub Actions deploy to GitHub Pages automatically on every push to main allowed the team to demo progress continuously.

5. **Leveraging team members' domain expertise accelerates delivery.** Having a teammate who already understood Battleship game mechanics meant the team could skip lengthy requirements elicitation and focus on implementation. In cross-functional teams, recognizing and utilizing each member's strengths — whether in game design, architecture, or frontend development — leads to more efficient task distribution and higher quality output.

14. Screenshots

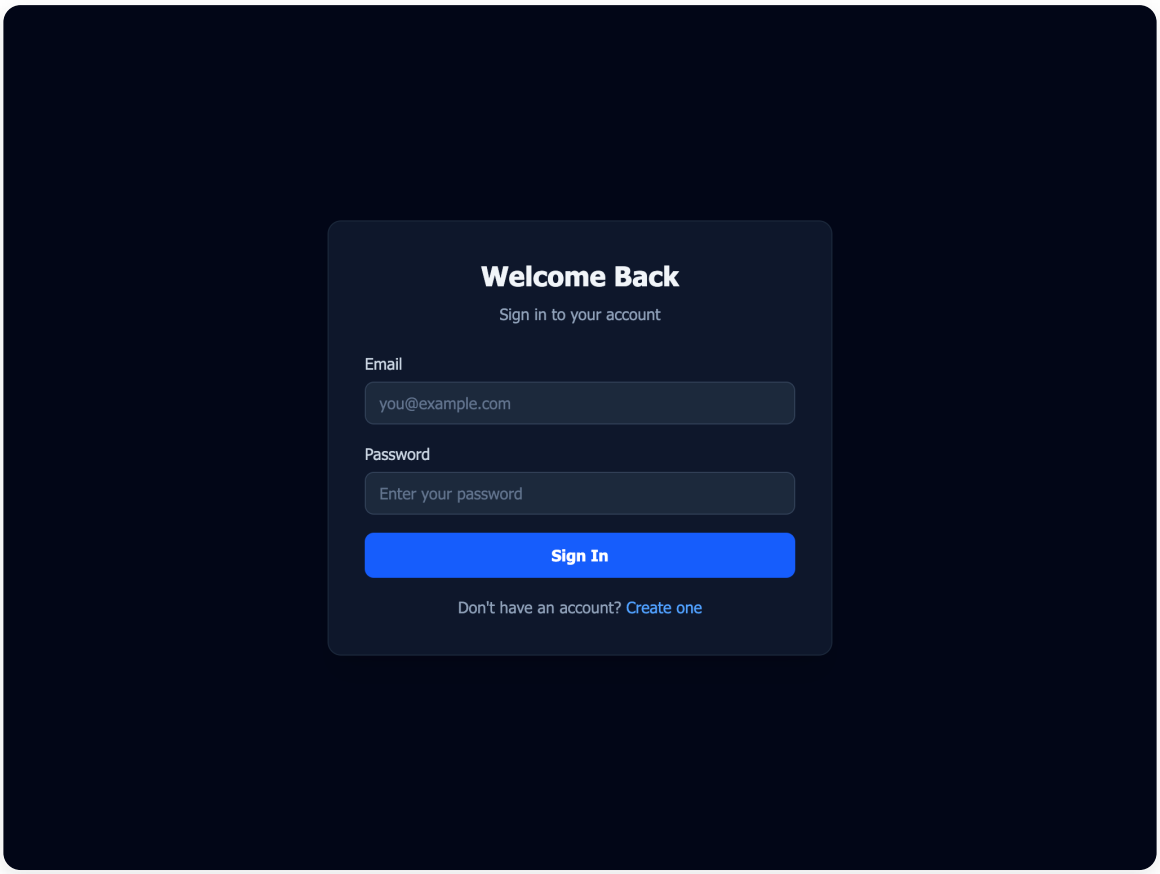
14.1 Landing Page



Landing Page

Dark navy blue theme with “Battleship Online” title, “Play Now” and “Sign In” call-to-action buttons.

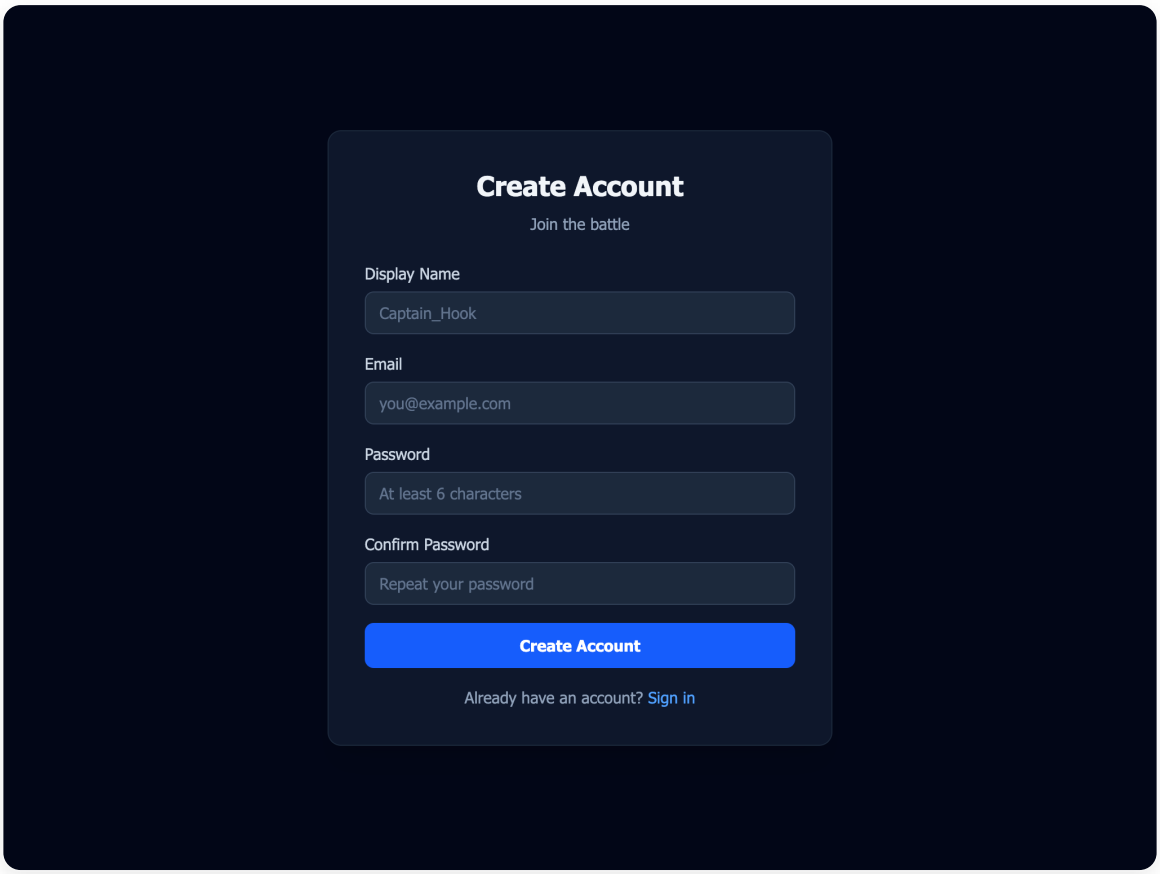
14.2 Login Page



Login Page

“Welcome Back” login form with email and password fields.

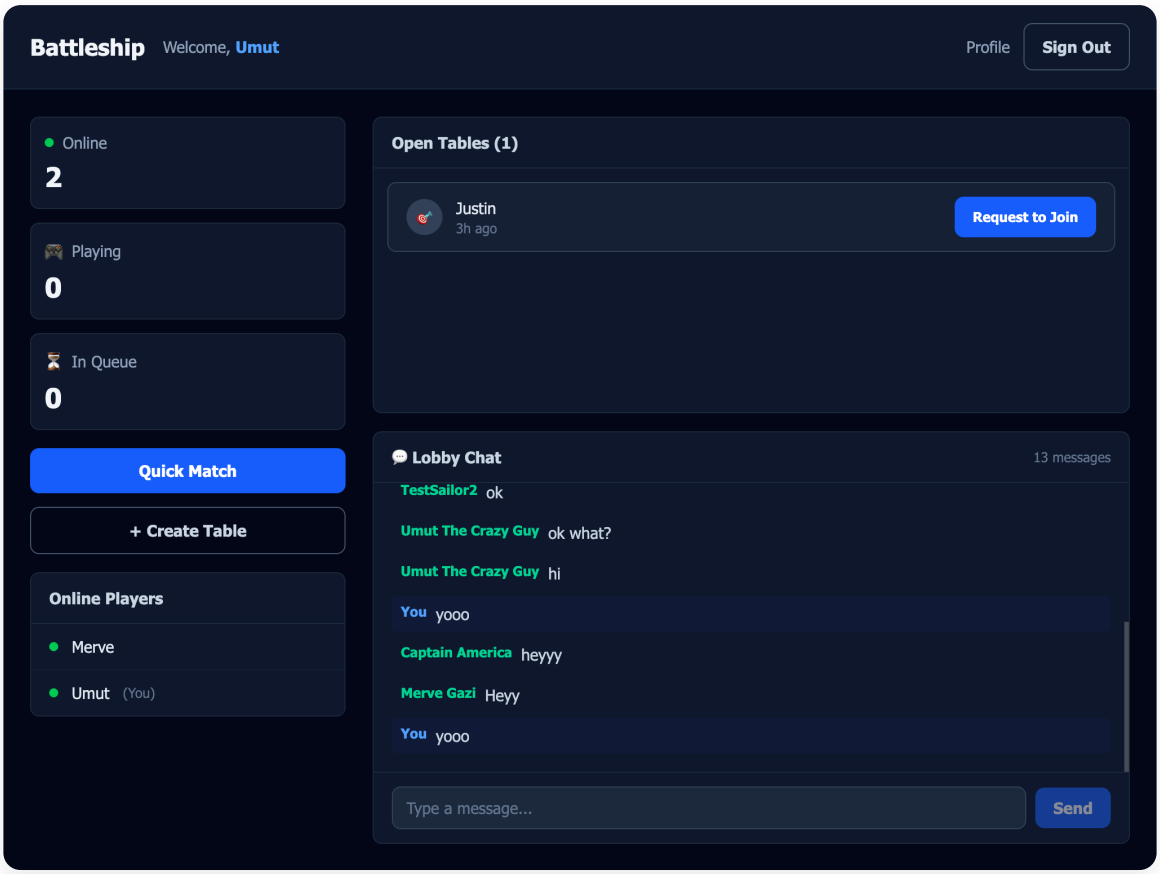
14.3 Register Page



Register Page

“Create Account” form with display name, email, password, and confirm password fields.

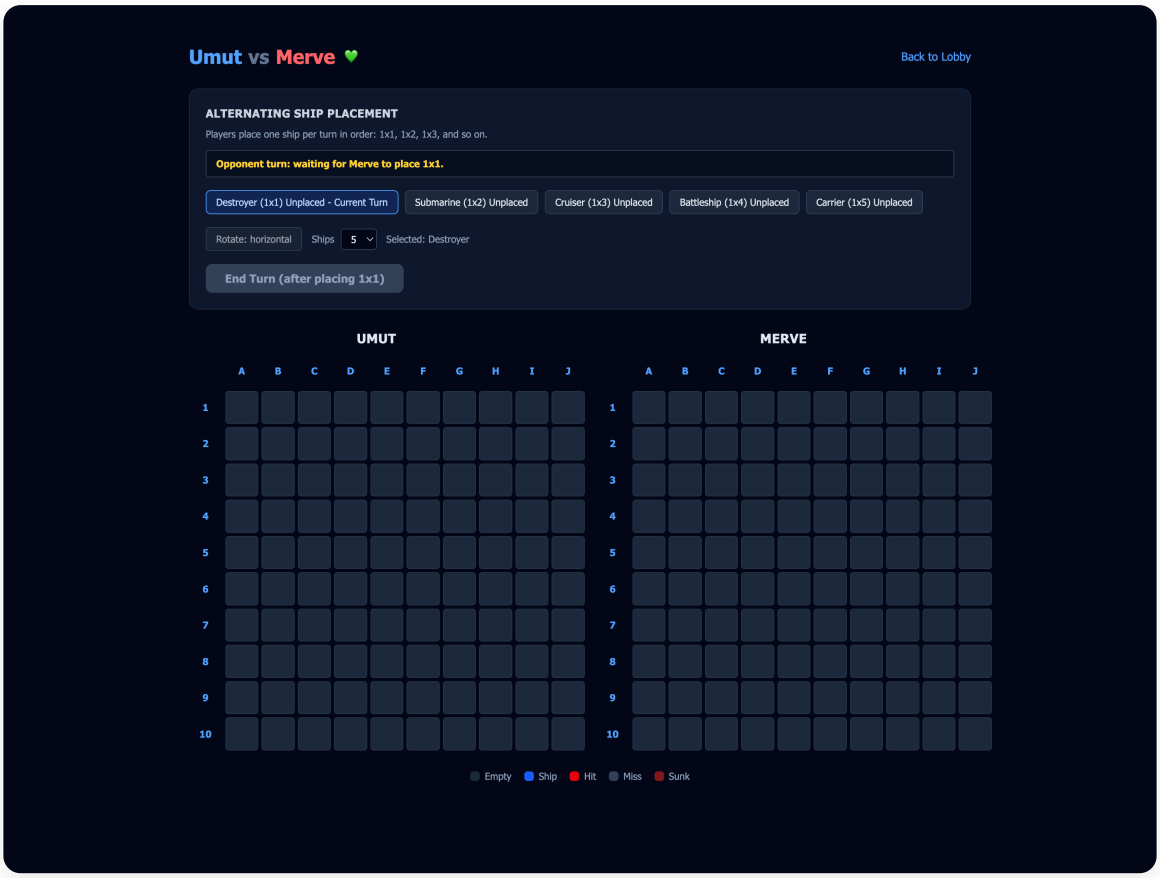
14.4 Lobby (Chat + Online Users + Tables)



Lobby

Full lobby view showing online player count, Quick Match button, open tables list, and real-time lobby chat panel.

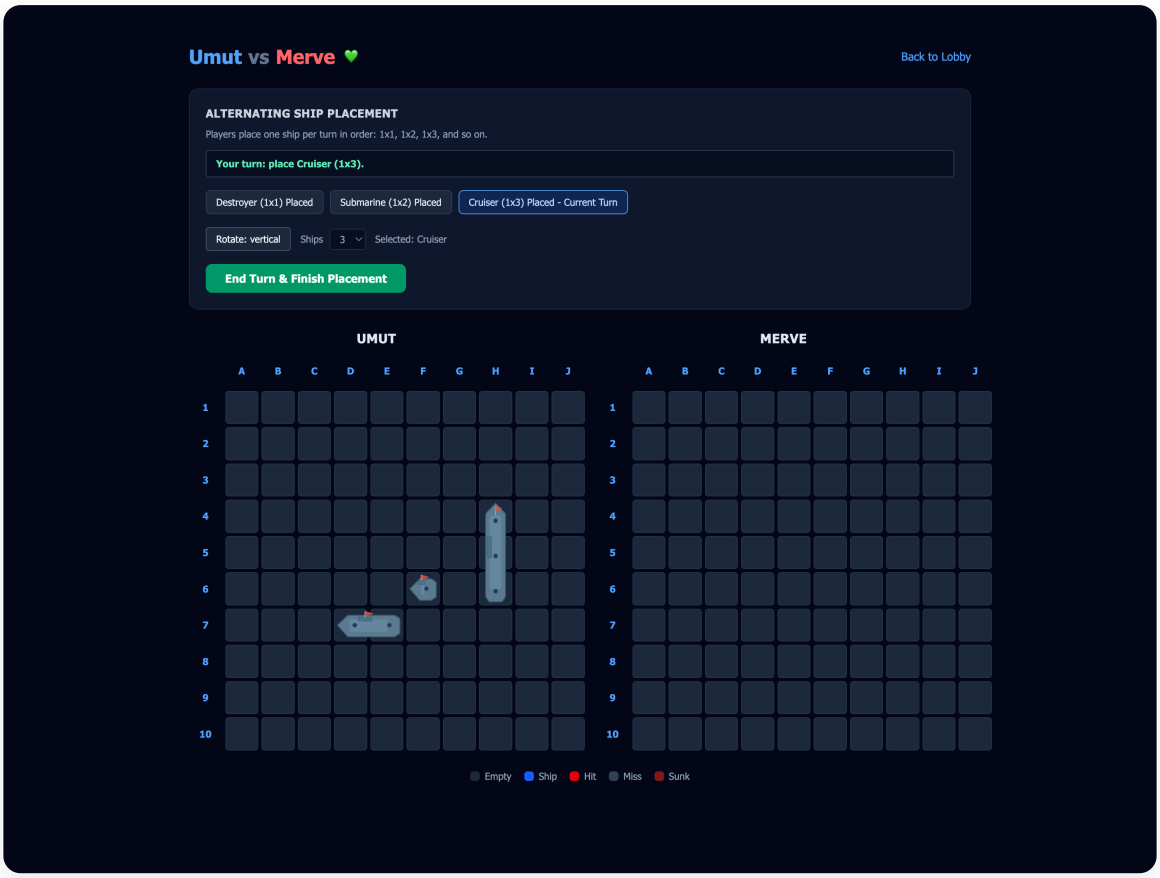
14.5 Ship Placement Phase (Empty Board)



Ship Placement - Empty

Game setup screen showing two empty 10×10 grids, alternating ship placement panel with all ships unplaced.

14.6 Ship Placement Phase (In Progress)



Ship Placement - In Progress

Ships being placed with SVG visuals — Destroyer and Submarine placed, Cruiser selected as current turn.

14.7 Game End — Victory Screen